

Three-body-ODE

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1 Introduction

The basic mathematics of circular motion. Consider a circle with radius r , circumference L , arc length S , and central angle θ , where θ is measured in radians.

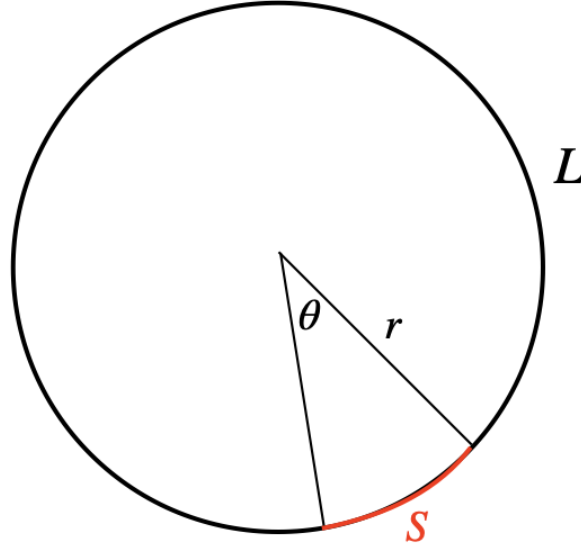


Figure 1: Relationship between the central angle and arc length in a circle.

1.1 Radian and Arc Length

The complete circumference is $L = 2\pi r$. Now suppose the central angle is θ , with corresponding arc length is S . Since $\frac{\theta}{2\pi} = \frac{S}{L}$, we have $S = \frac{\theta}{2\pi}L$. Substitute $L = 2\pi r$, we obtain $S = \frac{\theta}{2\pi}2\pi r = r\theta$.

1.2 Angular Velocity

For circular motion with a constant radius r , suppose the object moves through a small angle $\Delta\theta$. The corresponding change in arc length is $\Delta S = r\Delta\theta$. The tangential speed is defined as $v = \lim_{t \rightarrow 0} \frac{\Delta S}{\Delta t} = \lim_{t \rightarrow 0} \frac{r\Delta\theta}{\Delta t}$. Since the angular velocity is defined as $\omega = \frac{\Delta\theta}{\Delta t}$, we obtain $v = r\omega$.

1.3 Polar Unit Vectors

In two-dimensional polar coordinates, the position vector can be expressed as $\vec{r} = r\hat{r}$, where r is the distance from the origin and $\hat{r}$ is the radial unit vector. The tangential unit vector $\hat{\theta}$, which is perpendicular to $\hat{r}$ and points in the direction of increasing θ .

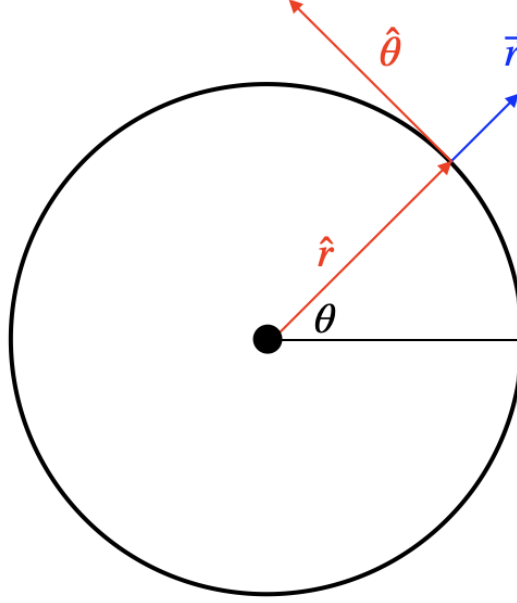


Figure 2: Radial and tangential unit vectors in polar coordinates

In Cartesian coordinates, the polar unit vectors can be expressed in terms of the Cartesian unit vectors $\hat{i}$ and $\hat{j}$ as $\hat{r} = \cos \theta \hat{i} + \sin \theta \hat{j}$ and $\hat{\theta} = -\sin \theta \hat{i} + \cos \theta \hat{j}$. Differentiate $\hat{r}$ with respect to θ :

$$\begin{aligned}\frac{d\hat{r}}{d\theta} &= \frac{d(\cos \theta \hat{i} + \sin \theta \hat{j})}{d\theta} = -\sin \theta \hat{i} + \cos \theta \hat{j} = \hat{\theta}, \\ \frac{d\hat{\theta}}{d\theta} &= \frac{d(-\sin \theta \hat{i} + \cos \theta \hat{j})}{d\theta} = -\cos \theta \hat{i} - \sin \theta \hat{j} = -\hat{r}.\end{aligned}$$

Therefore, the derivatives of the polar unit vectors with respect to θ are

$$\frac{d\hat{r}}{d\theta} = \hat{\theta} \quad \text{and} \quad \frac{d\hat{\theta}}{d\theta} = -\hat{r}.$$

1.4 Velocity in Polar Coordinates

The position vector is $\vec{r} = r\hat{r}$. Velocity is the time derivative of position $\vec{v} = \frac{d\vec{r}}{dt}$. By the product rule, we have $\vec{v} = \frac{d}{dt}(r\hat{r}) = \frac{dr}{dt}\hat{r} + r\frac{d\hat{r}}{dt}$. Then by the chain rule, $\frac{d\hat{r}}{dt} = \frac{d\hat{r}}{d\theta}\frac{d\theta}{dt} = \hat{\theta}\dot{\theta}$. Therefore, $\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}$.

The first component, $v_r = \dot{r}$, is the radial velocity, which describes how quickly the object moves toward or away from the origin. The second component, $r\dot{\theta}$, is the tangential (circumferential) velocity, which describes how quickly the object moves around the origin.

For a circular orbit, r is constant, which means $\dot{r} = 0$. Therefore, $\vec{v} = r\dot{\theta}\hat{\theta}$. Since $\omega = \dot{\theta}$, we recover $v = r\omega$.

1.5 Acceleration in Polar Coordinates

Acceleration is the time derivative of velocity $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}(\dot{r}\hat{r} + r\dot{\theta}\hat{\theta})$. In the first term, $\frac{d}{dt}(\dot{r}\hat{r}) = \ddot{r}\hat{r} + \dot{r}\frac{d\hat{r}}{dt} = \ddot{r}\hat{r} + \dot{r}\dot{\theta}\hat{\theta}$. In the second term, $\frac{d}{dt}(r\dot{\theta}\hat{\theta}) = \dot{r}\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta} + r\dot{\theta}\frac{d\hat{\theta}}{dt}$. Then by the chain rule, $\frac{d\hat{\theta}}{dt} = \frac{d\hat{\theta}}{d\theta}\frac{d\theta}{dt} = -\hat{r}\dot{\theta}$. Therefore, $\vec{a} = \ddot{r}\hat{r} + \dot{r}\dot{\theta}\hat{\theta} + \dot{r}\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta} - r\dot{\theta}^2\hat{r} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta}$.

Therefore, the acceleration can be decomposed into the radial and tangential components:

$$a_r = (\ddot{r} - r\dot{\theta}^2), \quad a_\theta = (2\dot{r}\dot{\theta} + r\ddot{\theta})$$

For a circular orbit, r is constant, which means $\dot{r} = 0$, $\ddot{r} = 0$. If the angular velocity is also constant, then $\dot{\theta} = \omega$, $\ddot{\theta} = 0$. Therefore, $\vec{a} = -r\omega^2\hat{r}$. The minus sign indicates that the acceleration points toward the center.

1.6 Centripetal Force

For uniform circular motion, we just obtained $\vec{a} = -r\omega^2\hat{r}$. According to Newton's second law, $\vec{F} = m\vec{a}$. Therefore, $\vec{F} = -mr\omega^2\hat{r}$. If we consider only the magnitude of the force,

$$\vec{F}_c = mr\omega^2.$$

Recall from 1.2 that $v = r\omega$. Therefore, $\omega = \frac{v}{r}$. Substitute this into the centripetal-force equation

$$\vec{F}_c = \frac{mv^2}{r}.$$

Gravitational force is defined as $F_g = G\frac{Mm}{r^2}$, where M and m are the two masses, r is the distance between them, and G is the gravitational constant. For the satellite example, gravity provides the force required for circular motion. Therefore, $F_c = F_g$. Thus $mr\omega^2 = G\frac{Mm}{r^2}$. Canceling m from both sides gives $r\omega^2 = G\frac{M}{r^2}$. Multiplying by R^2 , we obtain $r^3\omega^2 = GM$. Here, gravity acts as the centripetal force that keeps the satellite in circular motion.

2 The Three-Body Problem

2.1 Gravitational Force Between Two Bodies

Before considering three bodies, consider two masses m_1 and m_2 at positions r_1 and r_2 . The displacement vector pointing from body 1 to body 2 is $\vec{r}_{12} = \vec{r}_2 - \vec{r}_1$ and the distance between the two bodies is $r_{12} = \|\vec{r}_{12}\| = \|\vec{r}_2 - \vec{r}_1\|$.

For example, in two dimensions, $\vec{r}_1 = (x_1, y_1)$, $\vec{r}_2 = (x_2, y_2)$. So, $\vec{r}_{12} = (x_2 - x_1, y_2 - y_1)$ and $r_{12} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. The corresponding unit vector pointing from body 1 toward body 2 is $\hat{r}_{12} = \frac{\vec{r}_{12}}{r_{12}} = \frac{\vec{r}_2 - \vec{r}_1}{\|\vec{r}_2 - \vec{r}_1\|}$.

According to Newton's law of universal gravitation, the magnitude of the gravitational force between the two bodies is $\vec{F}_{12} = G\frac{m_1m_2}{r_{12}^2}\vec{r}_{12}$. Since the force acting on body 1 points toward body 2, its vector form is $\vec{F}_{12} = F_{12}\hat{r}_{12}$. Substitute the two equations: $\vec{F}_{12} = G\frac{m_1m_2}{r_{12}^2}\frac{\vec{r}_{12}}{r_{12}} = Gm_1m_2\frac{\vec{r}_{12}}{r_{12}^3}$. Thus, $\vec{F}_{12} = Gm_1m_2\frac{\vec{r}_2 - \vec{r}_1}{\|\vec{r}_2 - \vec{r}_1\|^3}$.

2.2 From Gravitational Force to an Equation of Motion

According to Newton's second law, $\vec{F} = m\vec{a}$. Therefore, for body 1, $m_1\vec{a}_1 = \vec{F}_{12}$. Substituting the gravitational force gives $m_1\vec{a}_1 = Gm_1m_2\frac{\vec{r}_2 - \vec{r}_1}{\|\vec{r}_2 - \vec{r}_1\|^3}$. We have, $\vec{a}_1 = Gm_2\frac{\vec{r}_2 - \vec{r}_1}{\|\vec{r}_2 - \vec{r}_1\|^3}$. Notice something interesting: m_1 disappears. The acceleration of body 1 due to body 2 depends on the mass m_2 , the positions of the two bodies, and the gravitational constant G .

Since the positions r_1 and r_2 change with time, they can be written as $r_1(t)$ and $r_2(t)$. Only their initial positions and velocities are known. The equation of motion determines how their positions evolve with time.

By definition, acceleration is $\vec{a}_1 = \frac{d^2\vec{r}_1}{dt^2}$. Therefore, $\frac{d^2\vec{r}_1}{dt^2} = Gm_2\frac{\vec{r}_2 - \vec{r}_1}{\|\vec{r}_2 - \vec{r}_1\|^3}$.

2.3 Initial Value Problem

We obtained

$$\frac{d^2 \vec{r}_1}{dt^2} = Gm_2 \frac{\vec{r}_2 - \vec{r}_1}{\|\vec{r}_2 - \vec{r}_1\|^3}.$$

The unknown is not simply $\vec{r}_1$. It is the function $\vec{r}_1(t)$ which describes the position of body 1 at every time t . Knowing the differential equation alone is not enough to determine a unique trajectory. Consider the much simpler equation $\frac{d^2 x}{dt^2} = 0$. Integrating once gives $\frac{dx}{dt} = C_1$, and integrating again gives $x(t) = C_1 t + C_2$. There are two unknown constants. To determine them, we need two pieces of information. In mechanics, these naturally correspond to $x(0) = x_0$ and $\dot{x} = v_0$. Then $x(t) = v_0 t + x_0$. So for a second-order equation of motion, we normally specify both the initial position and initial velocity.

For body 1, we need

$$\vec{r}_1(0) = \vec{r}_{1,0}, \quad \dot{\vec{r}}_1(0) = \vec{v}_{1,0}.$$

Similarly for body 2, we need

$$\vec{r}_2(0) = \vec{r}_{2,0}, \quad \dot{\vec{r}}_2(0) = \vec{v}_{2,0}.$$

This combination of **differential equations + initial conditions** is called an Initial Value Problem (IVP).

2.4 Equations of Motion for Three Bodies

Consider three bodies with masses m_1, m_2, m_3 and positions $\vec{r}_1(t), \vec{r}_2(t), \vec{r}_3(t)$. For body 1, there are now two gravitational forces acting on it: $\vec{F}_{12}$ from body2 and $\vec{F}_{13}$ from body3. By the principle of superposition, $\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13}$. Therefore, $\vec{F}_1 = Gm_1 m_2 \frac{\vec{r}_2 - \vec{r}_1}{\|\vec{r}_2 - \vec{r}_1\|^3} + Gm_1 m_3 \frac{\vec{r}_3 - \vec{r}_1}{\|\vec{r}_3 - \vec{r}_1\|^3}$. Using Newton's second law, we obtain

$$\frac{d^2 \vec{r}_1}{dt^2} = Gm_2 \frac{\vec{r}_2 - \vec{r}_1}{\|\vec{r}_2 - \vec{r}_1\|^3} + Gm_3 \frac{\vec{r}_3 - \vec{r}_1}{\|\vec{r}_3 - \vec{r}_1\|^3}.$$

Exactly the same reasoning gives

$$\frac{d^2 \vec{r}_2}{dt^2} = Gm_1 \frac{\vec{r}_1 - \vec{r}_2}{\|\vec{r}_1 - \vec{r}_2\|^3} + Gm_3 \frac{\vec{r}_3 - \vec{r}_2}{\|\vec{r}_3 - \vec{r}_2\|^3},$$

$$\frac{d^2 \vec{r}_3}{dt^2} = Gm_1 \frac{\vec{r}_1 - \vec{r}_3}{\|\vec{r}_1 - \vec{r}_3\|^3} + Gm_2 \frac{\vec{r}_2 - \vec{r}_3}{\|\vec{r}_2 - \vec{r}_3\|^3}.$$

Now we have the three-body equations of motion.

In two dimensions, each position vector contains two components, $\vec{r}_i(t) = (x_i(t), y_i(t))$, so the three vector equations represent six coupled second-order ODEs.

2.5 Converting Second-Order ODEs to First-Order ODEs

For each body, $\vec{v}_i(t) = \frac{d\vec{r}_i}{dt}$. And because acceleration is the derivative of velocity, $\frac{d\vec{v}_i(t)}{dt} = \frac{d^2\vec{r}_i}{dt^2}$. Therefore, the three second-order equations become six vector first-order equations:

$$\begin{aligned}\frac{d\vec{r}_1}{dt} &= \vec{v}_1, & \frac{d\vec{v}_1}{dt} &= Gm_2 \frac{\vec{r}_2 - \vec{r}_1}{\|\vec{r}_2 - \vec{r}_1\|^3} + Gm_3 \frac{\vec{r}_3 - \vec{r}_1}{\|\vec{r}_3 - \vec{r}_1\|^3} \\ \frac{d\vec{r}_2}{dt} &= \vec{v}_2, & \frac{d\vec{v}_2}{dt} &= Gm_1 \frac{\vec{r}_1 - \vec{r}_2}{\|\vec{r}_1 - \vec{r}_2\|^3} + Gm_3 \frac{\vec{r}_3 - \vec{r}_2}{\|\vec{r}_3 - \vec{r}_2\|^3} \\ \frac{d\vec{r}_3}{dt} &= \vec{v}_3, & \frac{d\vec{v}_3}{dt} &= Gm_1 \frac{\vec{r}_1 - \vec{r}_3}{\|\vec{r}_1 - \vec{r}_3\|^3} + Gm_2 \frac{\vec{r}_2 - \vec{r}_3}{\|\vec{r}_2 - \vec{r}_3\|^3}\end{aligned}$$

In two dimensions, each position and velocity vector contains two components,

$$\vec{r}_i(t) = (x_i(t), y_i(t)) \quad \vec{v}_i(t) = (v_{xi}(t), v_{yi}(t)),$$

The distances between the three pairs of bodies are defined as r_{12} , r_{13} , and r_{23} :

$$\begin{aligned}r_{12} &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}, \\ r_{13} &= \sqrt{(x_3 - x_1)^2 + (y_3 - y_1)^2}, \\ r_{23} &= \sqrt{(x_3 - x_2)^2 + (y_3 - y_2)^2}.\end{aligned}$$

Thus, the twelve first-order ODEs are

$$\begin{aligned}\dot{x}_1 &= v_{x1}, \\ \dot{y}_1 &= v_{y1}, \\ \dot{x}_2 &= v_{x2}, \\ \dot{y}_2 &= v_{y2}, \\ \dot{x}_3 &= v_{x3}, \\ \dot{y}_3 &= v_{y3}, \\ \dot{v}_{x1} &= Gm_2 \frac{x_2 - x_1}{r_{12}^3} + Gm_3 \frac{x_3 - x_1}{r_{13}^3}, \\ \dot{v}_{y1} &= Gm_2 \frac{y_2 - y_1}{r_{12}^3} + Gm_3 \frac{y_3 - y_1}{r_{13}^3}, \\ \dot{v}_{x2} &= Gm_1 \frac{x_1 - x_2}{r_{12}^3} + Gm_3 \frac{x_3 - x_2}{r_{23}^3}, \\ \dot{v}_{y2} &= Gm_1 \frac{y_1 - y_2}{r_{12}^3} + Gm_3 \frac{y_3 - y_2}{r_{23}^3}, \\ \dot{v}_{x3} &= Gm_1 \frac{x_1 - x_3}{r_{13}^3} + Gm_2 \frac{x_2 - x_3}{r_{23}^3}, \\ \dot{v}_{y3} &= Gm_1 \frac{y_1 - y_3}{r_{13}^3} + Gm_2 \frac{y_2 - y_3}{r_{23}^3}.\end{aligned}$$

2.6 Figure-Eight Orbit

This is a periodic symmetric orbit where the particles follow a figure eight trajectory in the plane. The orbit is such that the three masses chase one another around a single figure-eight shaped curve. The initial values for the periodic three-body figure-eight orbit, computed numerically by Carles Simó, place the three equal masses at specific positions and velocities.

System Constants: $m_1 = m_2 = m_3 = 1$, and $G = 1$

Orbital Period: $T \approx 6.32591398$

Initial Positions:

$$\vec{r}_1(0) = (-0.97000436, 0.24308753),$$

$$\vec{r}_2(0) = (0.97000436, -0.24308753),$$

$$\vec{r}_3(0) = (0.0, 0.0).$$

Initial Velocities:

$$\vec{v}_1(0) = (0.466203685, 0.432365730),$$

$$\vec{v}_2(0) = (0.466203685, 0.432365730),$$

$$\vec{v}_3(0) = (-0.93240737, -0.86473146).$$